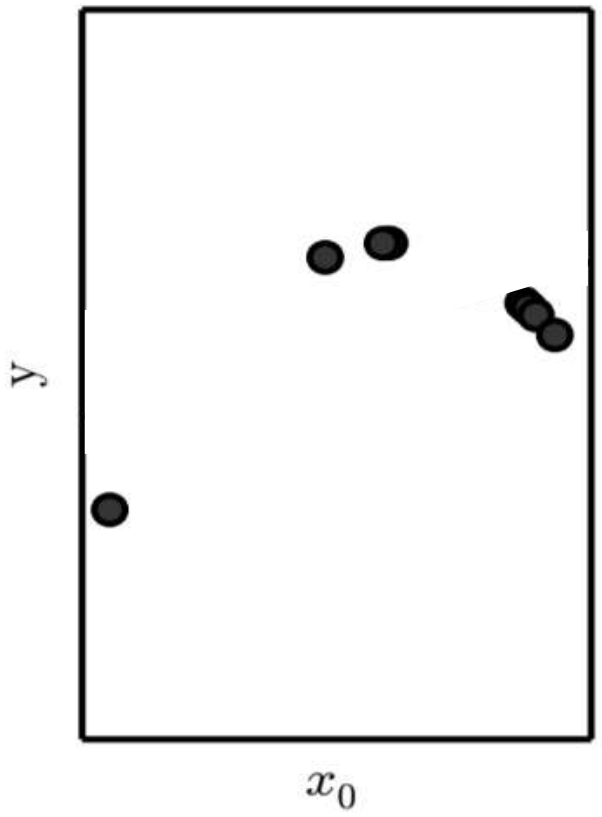
Regularization

Underfitting and Overfitting in Polynomial Estimation



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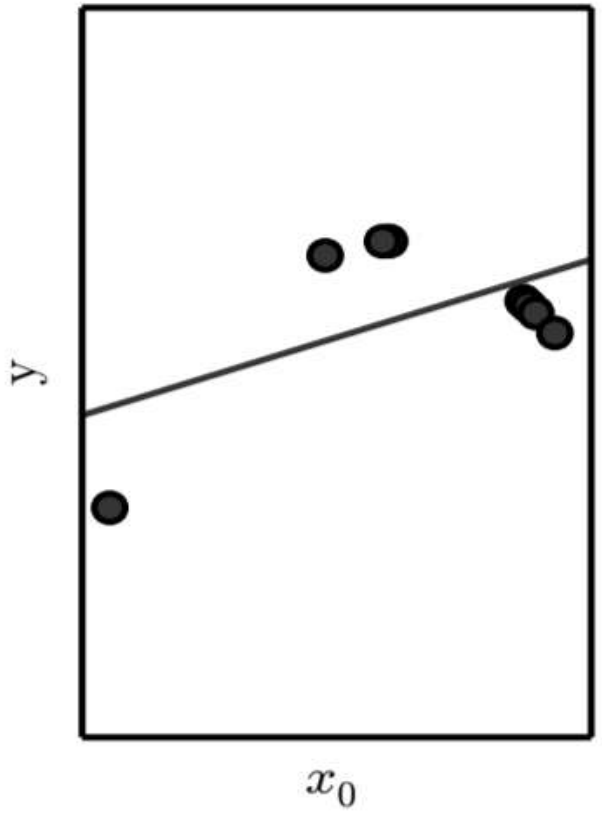

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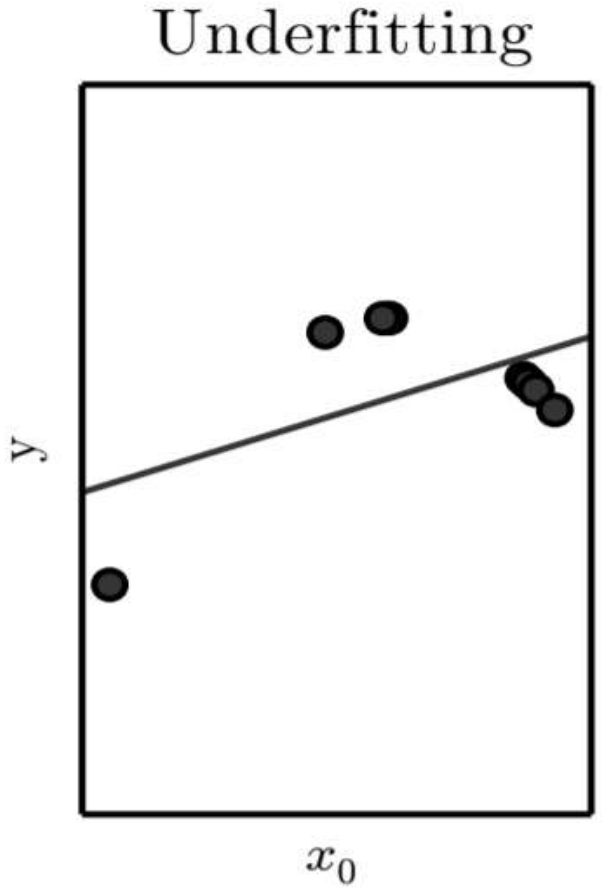

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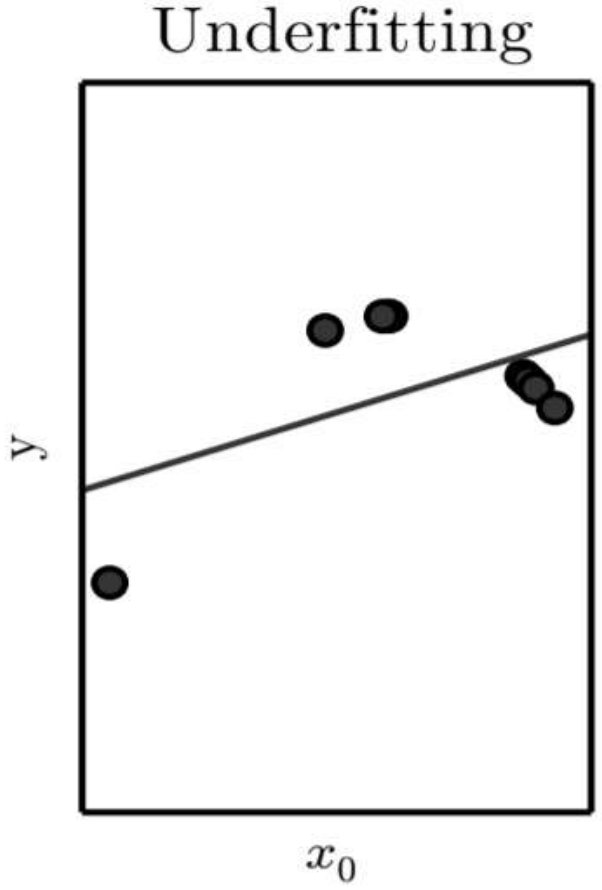

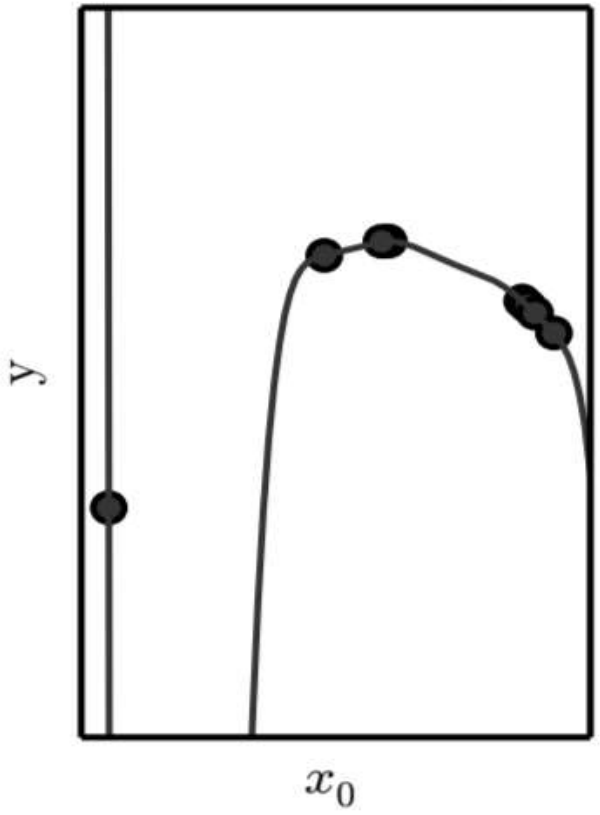

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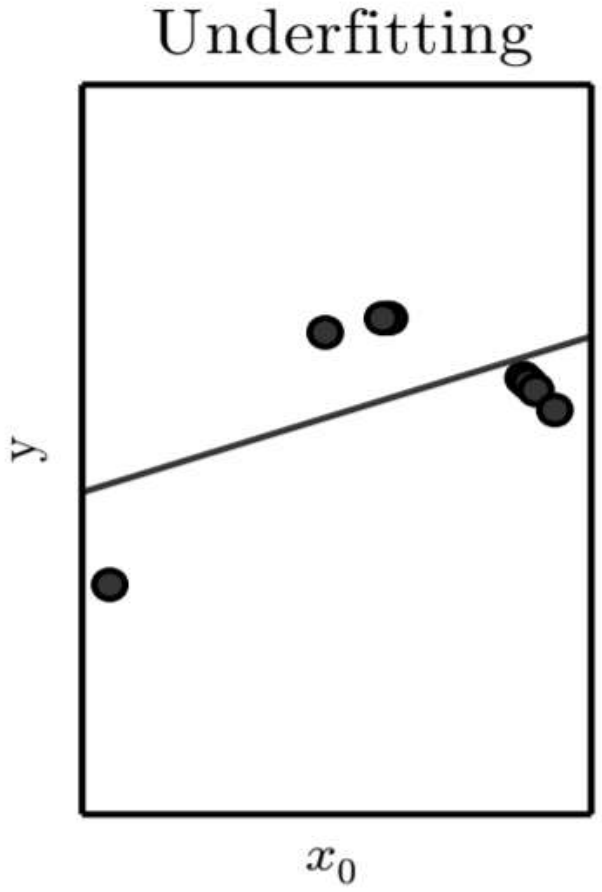

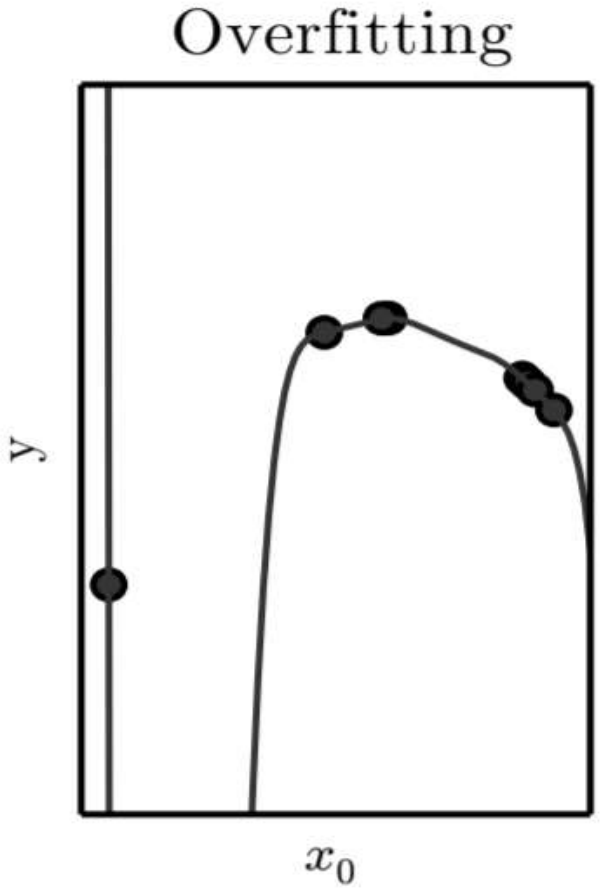

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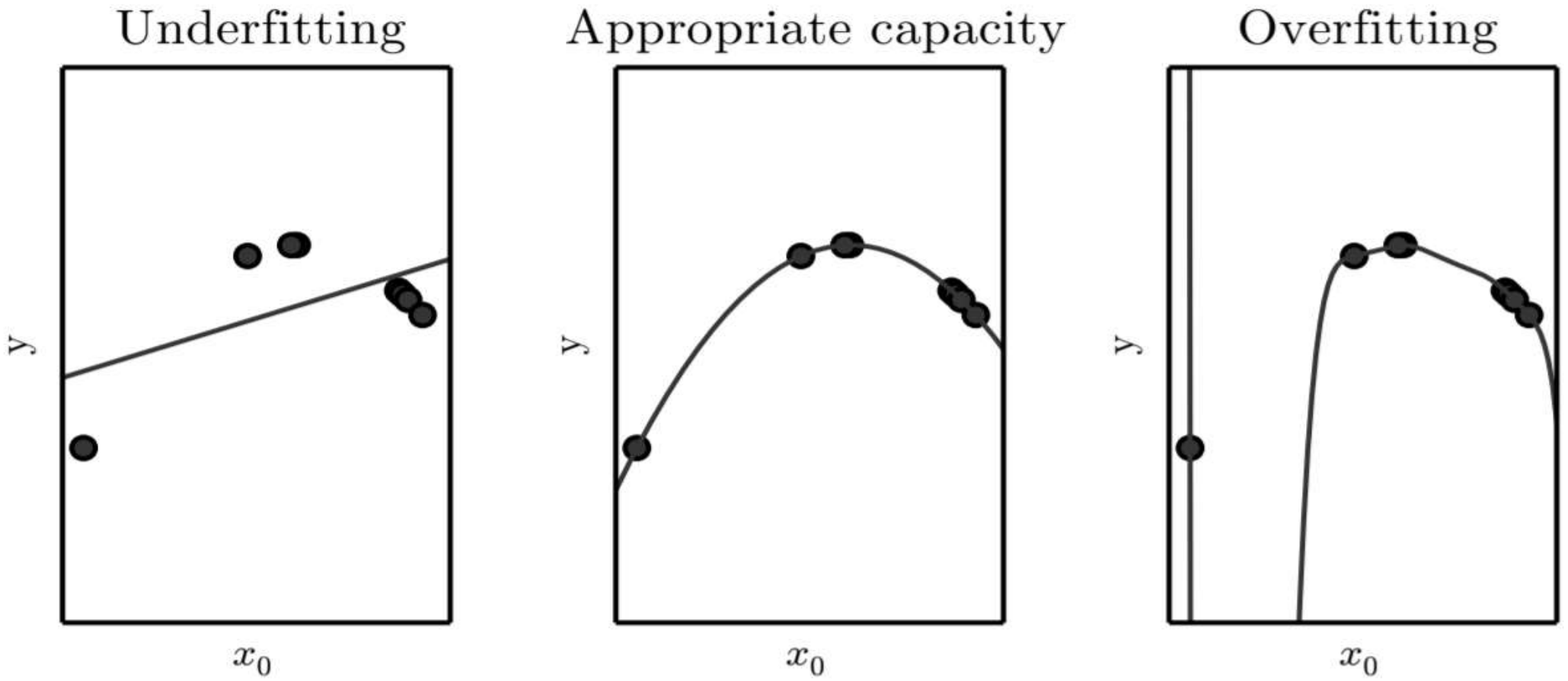

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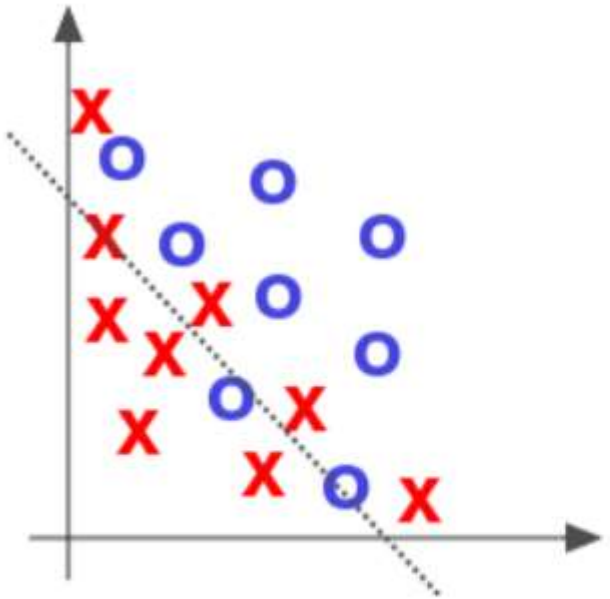
Over- and Underfitting



Underfitted

Source: Deep Learning by Adam Gibson, Josh Patterson, O'Reily Media Inc., 2017

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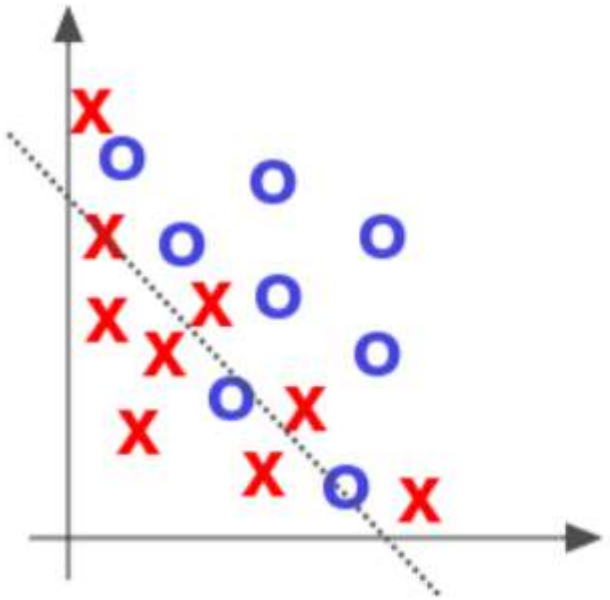

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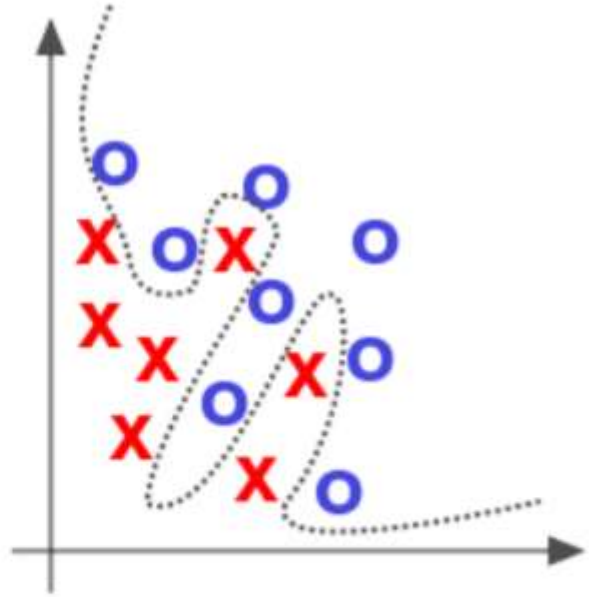

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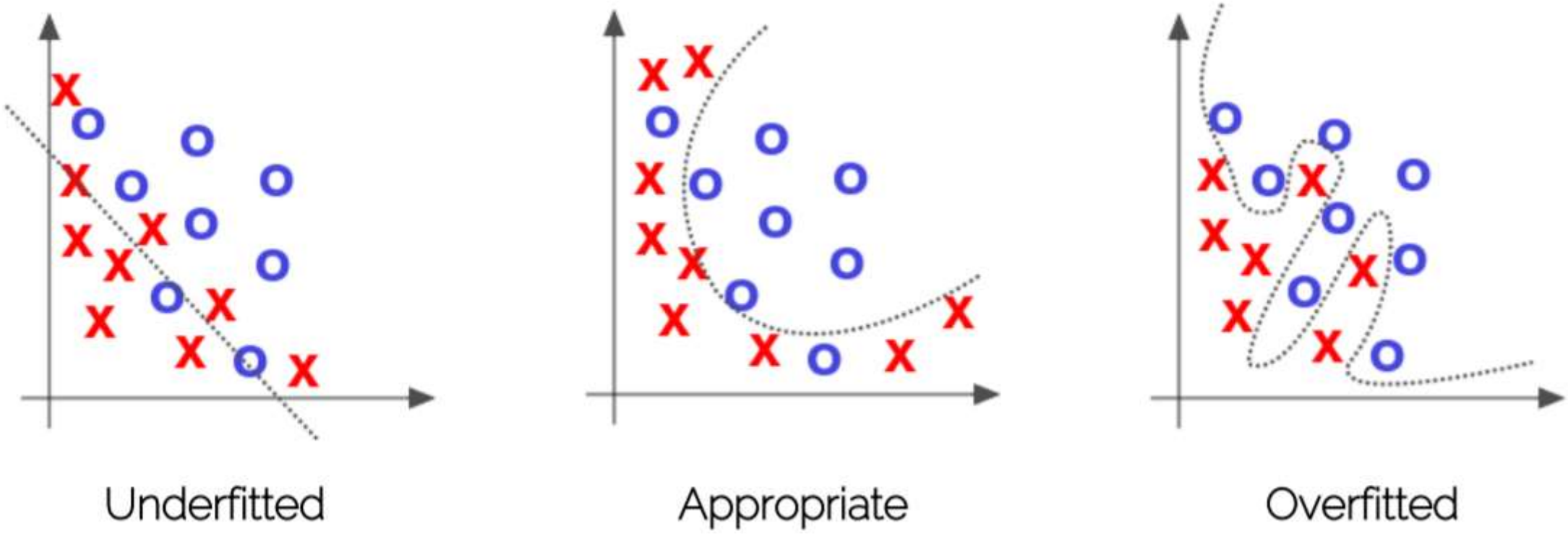

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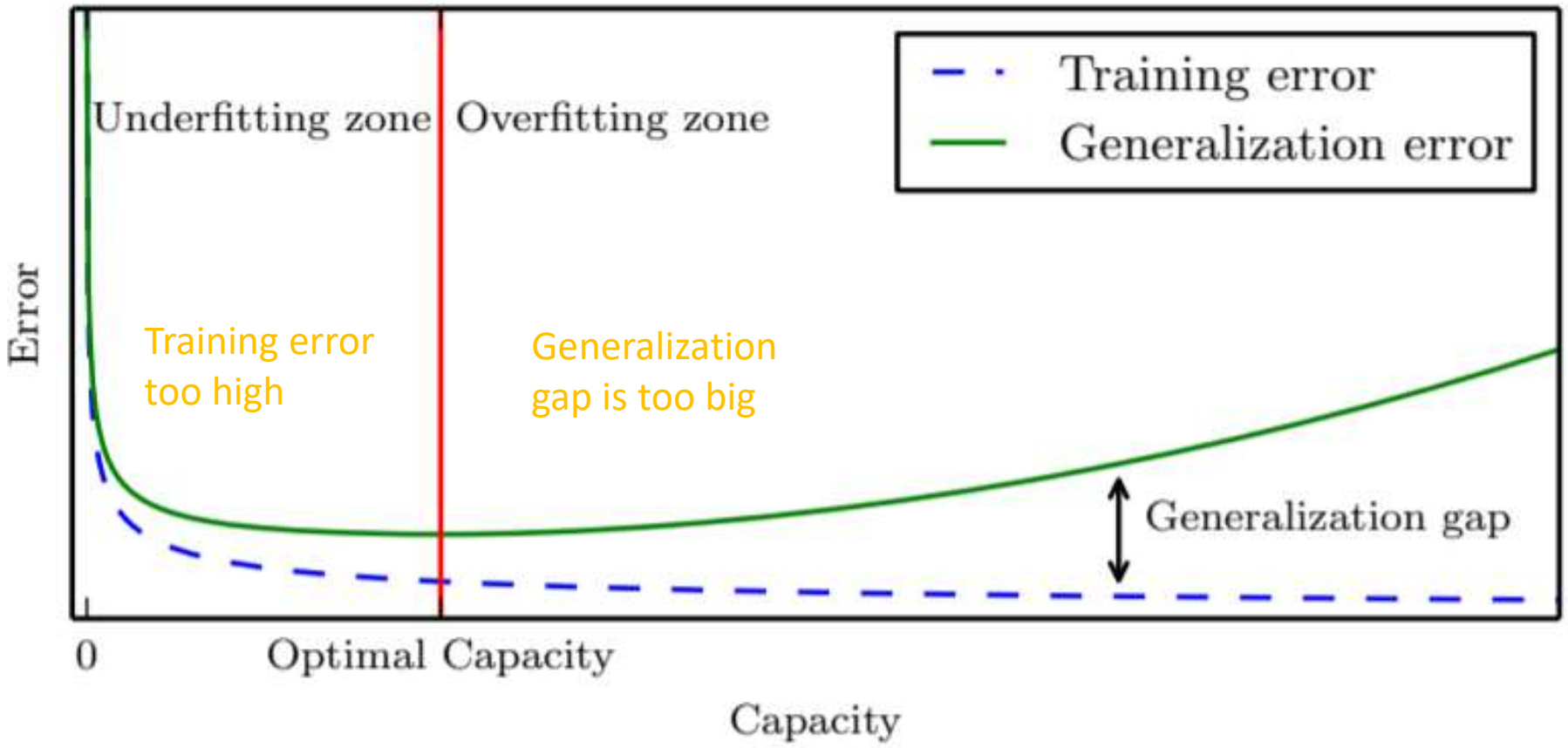
Training a Neural Network

Training/ Validation curve



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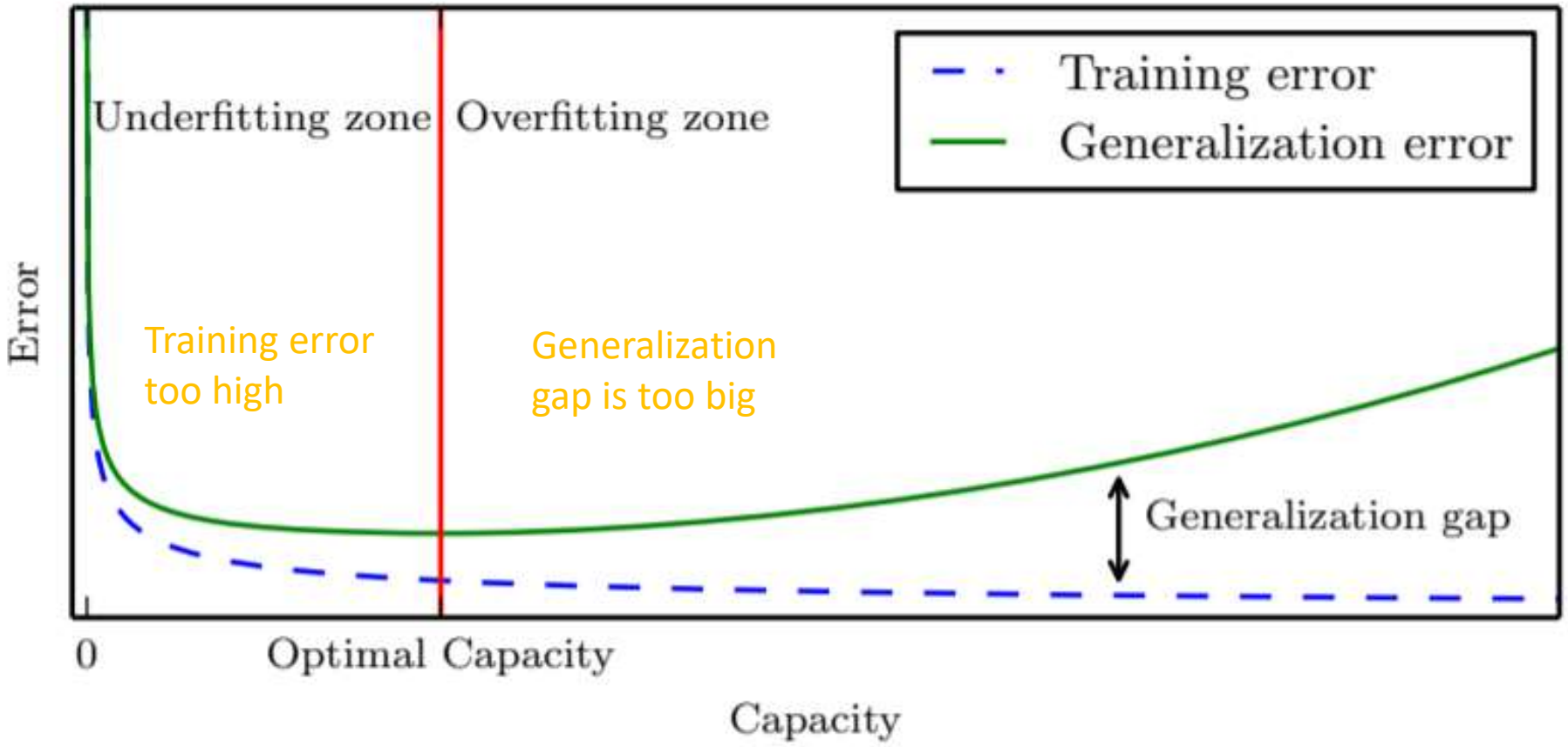

How can we prevent our model from overfitting?

Source: Ian Goodfellow , Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press

Training a Neural Network

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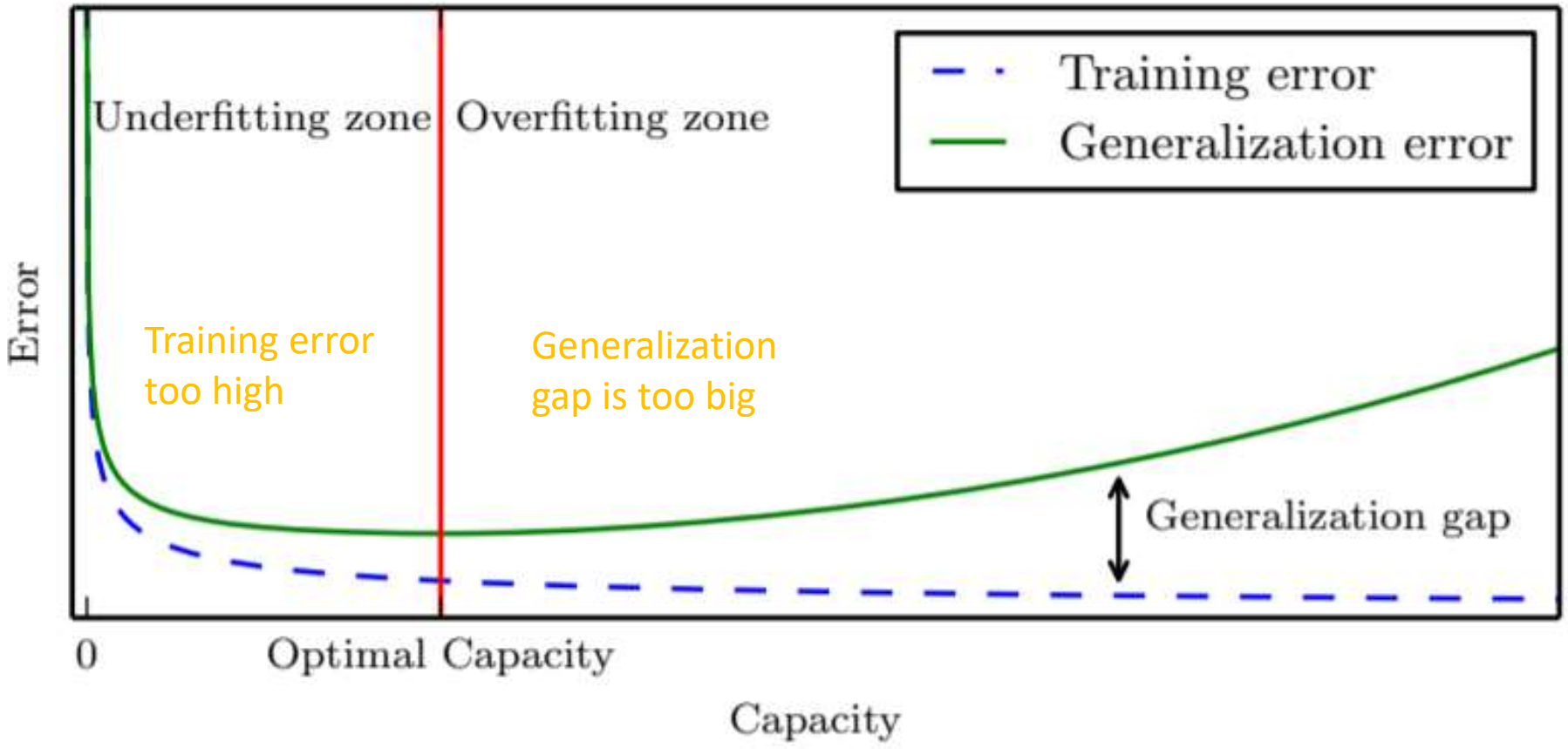

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Regularization

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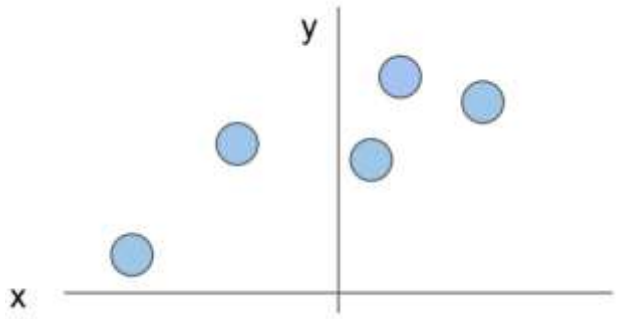
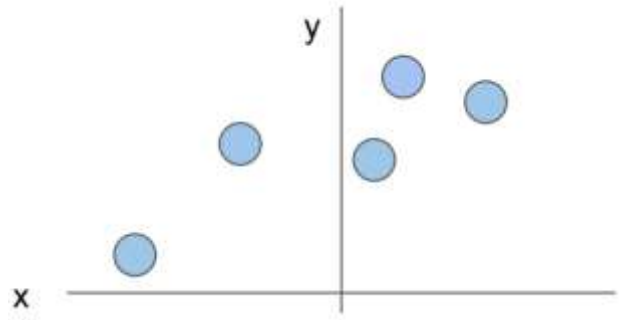
Regularization: Prefer Simpler Models



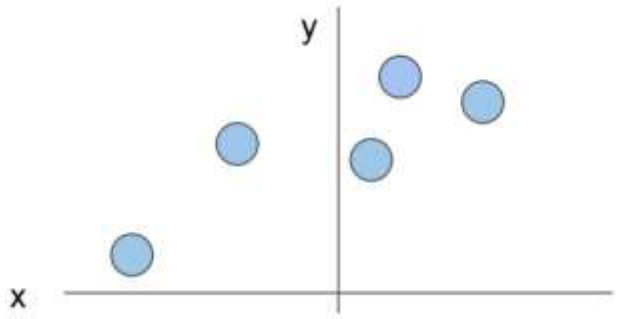
Regularization: Prefer Simpler Models



$$L(W) = \underbrace{\frac{1}{N}\sum_{i=1}^{N} L_i(f(x_i, W), y_i)}$$

Data loss: Model predictions should match training data

Regularization: Prefer Simpler Models



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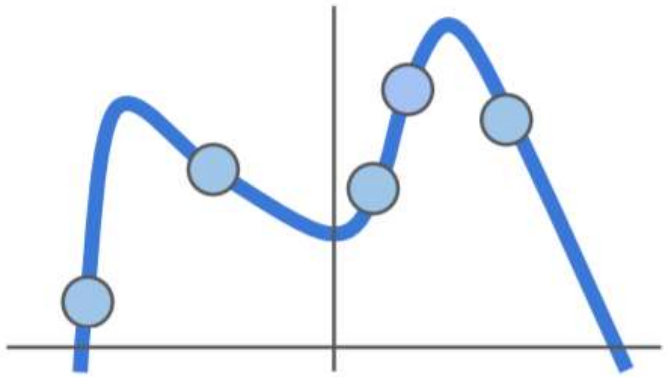

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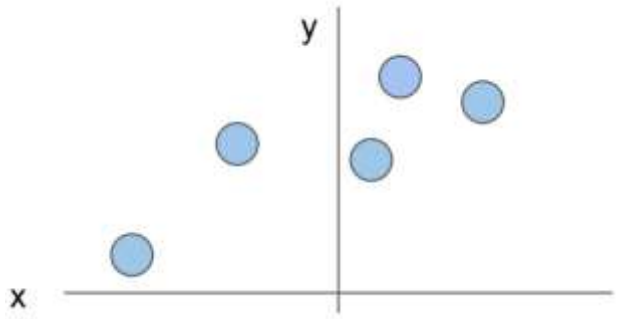

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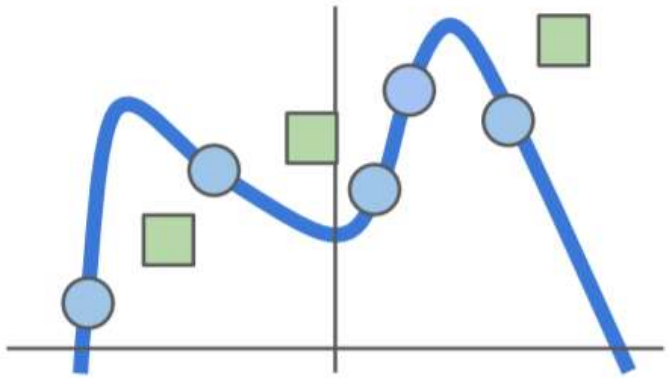

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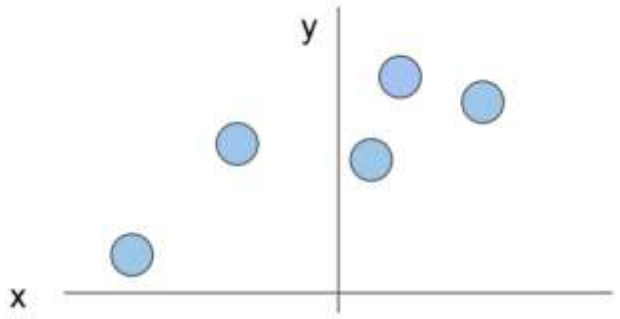

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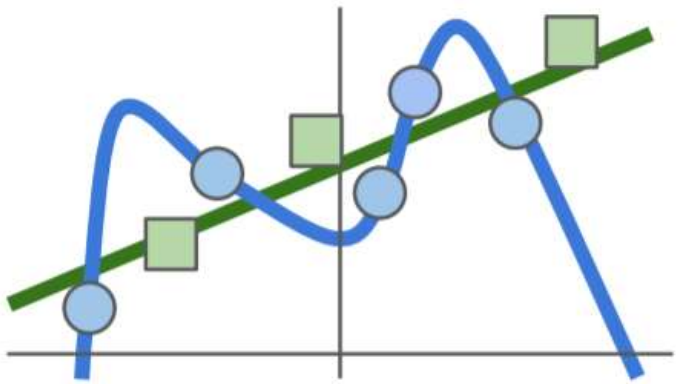

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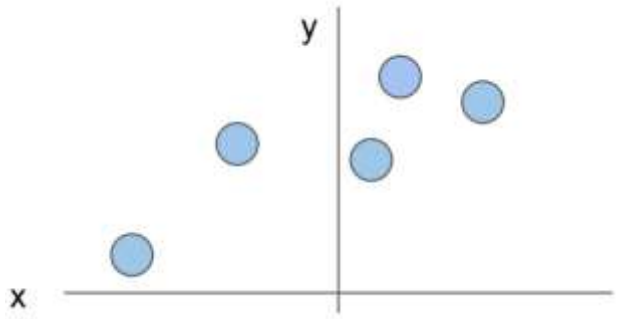

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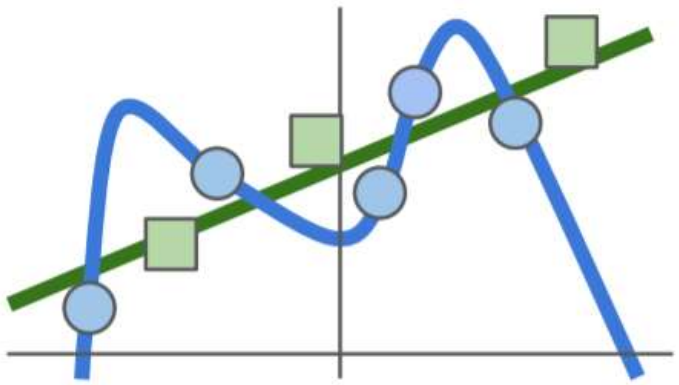
Data loss: Model predictions should match training data

Regularization: Prevent the model from doing *too* well on training data



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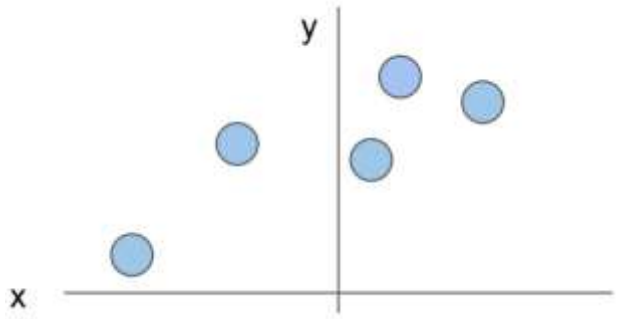

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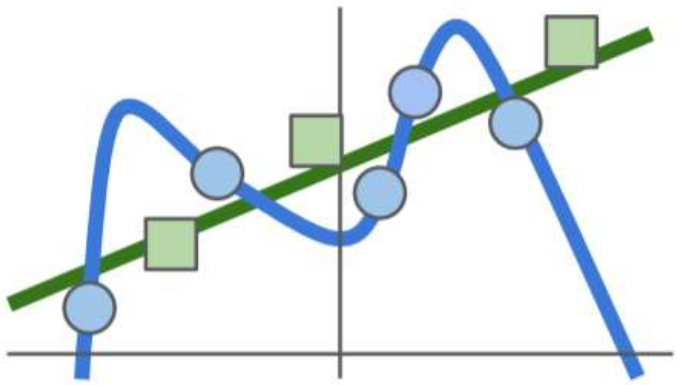

Occam's Razor:
"Among competing hypotheses, the simplest is the best"
William of Ockham, 1285 - 1347

Regularization

- Loss function $L(\boldsymbol{y}, \hat{\boldsymbol{y}}, \boldsymbol{\theta}) = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 + \lambda R(\boldsymbol{\theta})$

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization

- Loss function $L(\boldsymbol{y}, \hat{\boldsymbol{y}}, \boldsymbol{\theta}) = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 + \lambda R(\boldsymbol{\theta})$

- Regularization techniques
  - L2 regularization
  - L1 regularization

  Add **regularization term** to loss function

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- Regularization techniques
  - L2 regularization
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  Add **regularization term** to loss function

  - Max norm regularization
  - Dropout
  - Early stopping
  - ...

  More details later

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization: Example

- Input: 3 features $\boldsymbol{x}$ = [1, 2, 1]

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Regularization: Example

- Input: 3 features $\boldsymbol{x}$ = [1, 2, 1]
- Two linear classifiers that give the same result:
- θ_1 = [0, 0.75, 0]
- θ_2 = [0.25, 0.5, 0.25]

Regularization: Example

- Input: 3 features $\boldsymbol{x}$ = [1, 2, 1]
- Two linear classifiers that give the same result:
- θ_1 = [0, 0.75, 0] Ignores 2 features
- θ_2 = [0.25, 0.5, 0.25] Takes information from all features

Regularization: Example

- Loss $L(\boldsymbol{y}, \hat{\boldsymbol{y}}, \boldsymbol{\theta}) = \sum_{i=1}^{n} \left(x_i \theta_{ji} - y_i\right)^2 + \lambda R(\boldsymbol{\theta})$

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Regularization: Example

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$$x = [1, 2, 1], \theta_1 = [0, 0.75, 0], \theta_2 = [0.25, 0.5, 0.25]$$

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization: Example

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- L2 regularization $\boldsymbol{R(\theta)} = \sum_{i=1}^{n} \theta_i^2$

$\theta_1 \longrightarrow 0 + 0.75^2 + 0 = 0.5625$

$\theta_2 \longrightarrow 0.25^2 + 0.5^2 + 0.25^2 = 0.375$ Minimization

$x = [1, 2, 1], \theta_1 = [0, 0.75, 0], \theta_2 = [0.25, 0.5, 0.25]$

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization: Example

- Loss $L(\boldsymbol{y}, \hat{\boldsymbol{y}}, \boldsymbol{\theta}) = \sum_{i=1}^{n} \left(x_i \theta_{ji} - y_i\right)^2 + \lambda R(\boldsymbol{\theta})$
- L1 regularization $R(\boldsymbol{\theta}) = \sum_{i=1}^{n} |\theta_i|$

$\theta_1 \longrightarrow 0 + 0.75 + 0 = 0.75$

$\theta_2 \longrightarrow 0.25 + 0.5 + 0.25 = 1$

$x = [1, 2, 1], \theta_1 = [0, 0.75, 0], \theta_2 = [0.25, 0.5, 0.25]$

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization: Example

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$\boldsymbol{\theta_1 = [0, 0.75, 0]}$ → L1 regularization enforces **sparsity**

Regularization: Example

- Input: 3 features $\boldsymbol{x} = [1, 2, 1]$

- Two linear classifiers that give the same result:

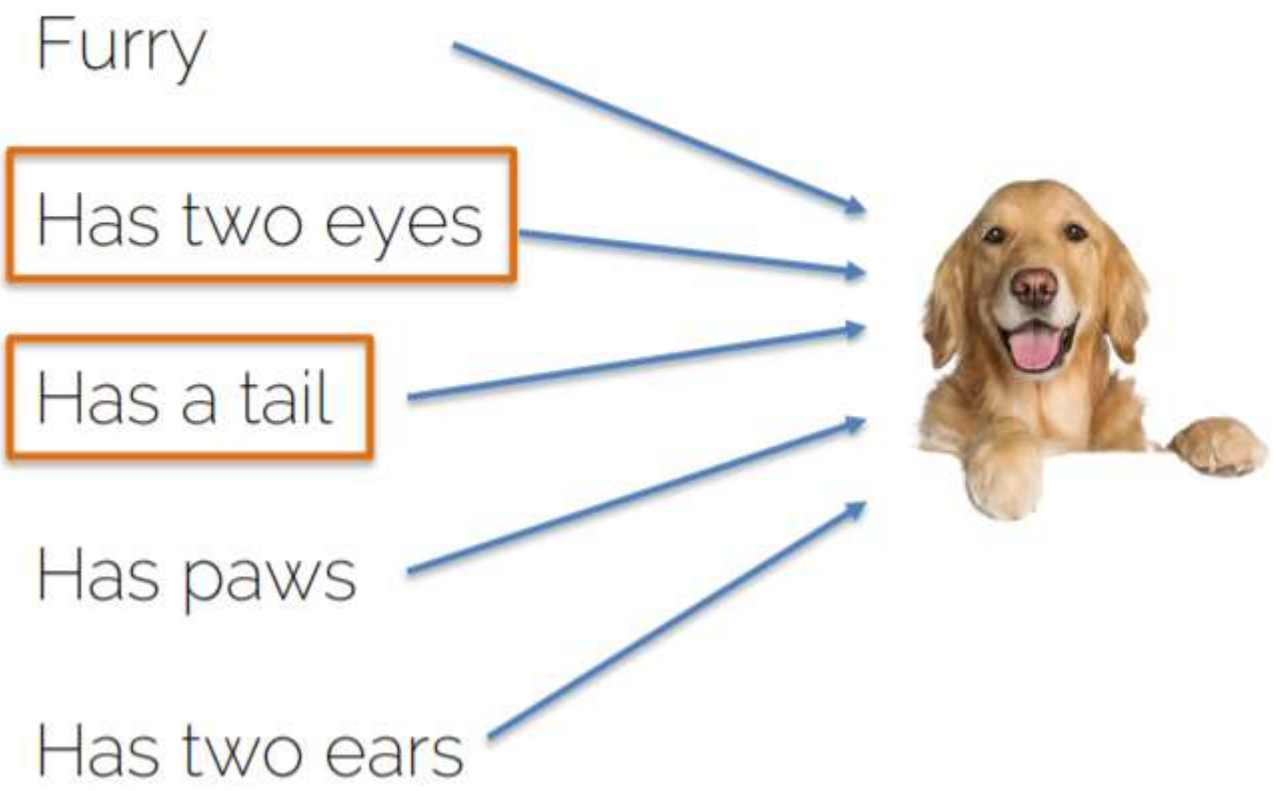
$\theta_1 = [0, 0.75, 0]$ → L1 regularization enforces **sparsity**

$\theta_2 = [0.25, 0.5, 0.25]$ → L2 regularization enforces that the weights have **similar values**

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization: Effect

Dog classifier takes different inputs



L1 regularization will focus all the attention to a few key features

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization: Effect

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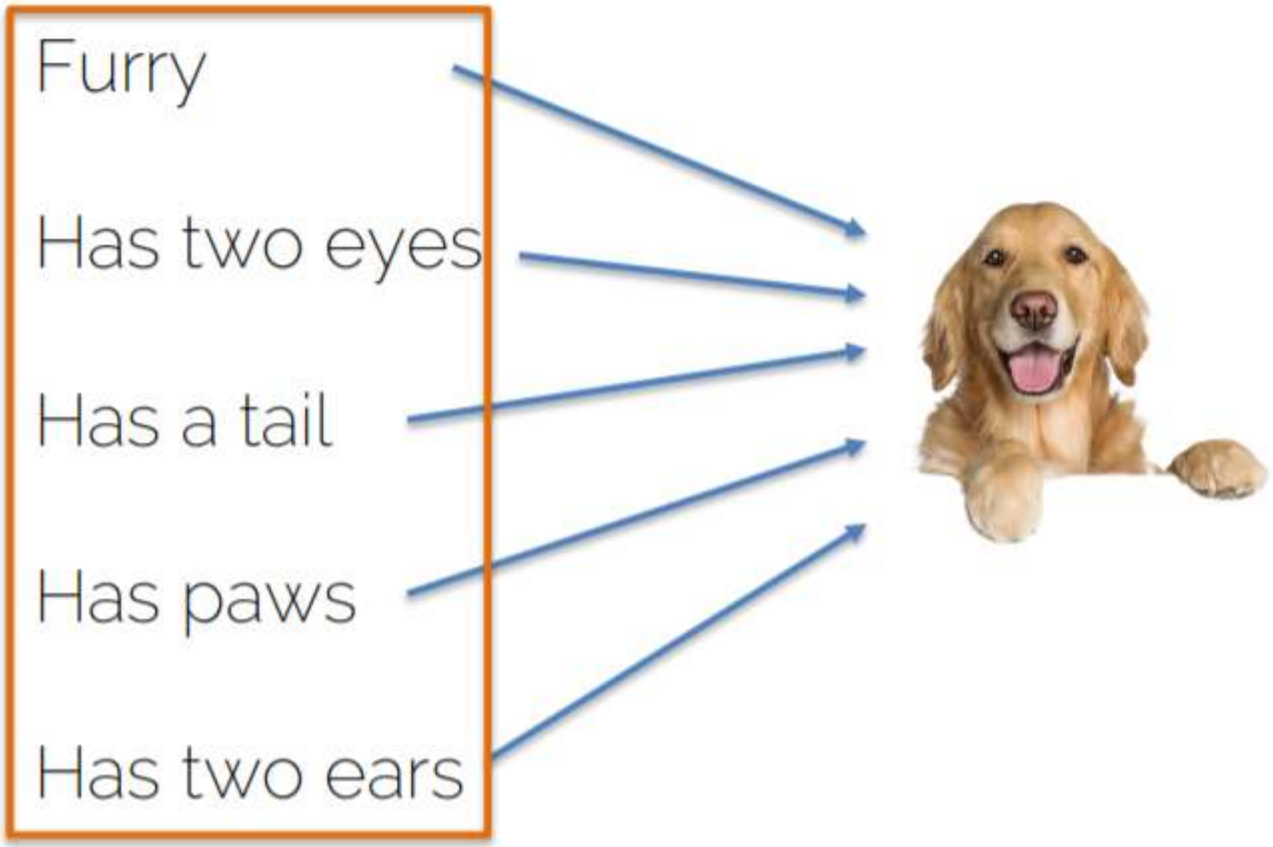
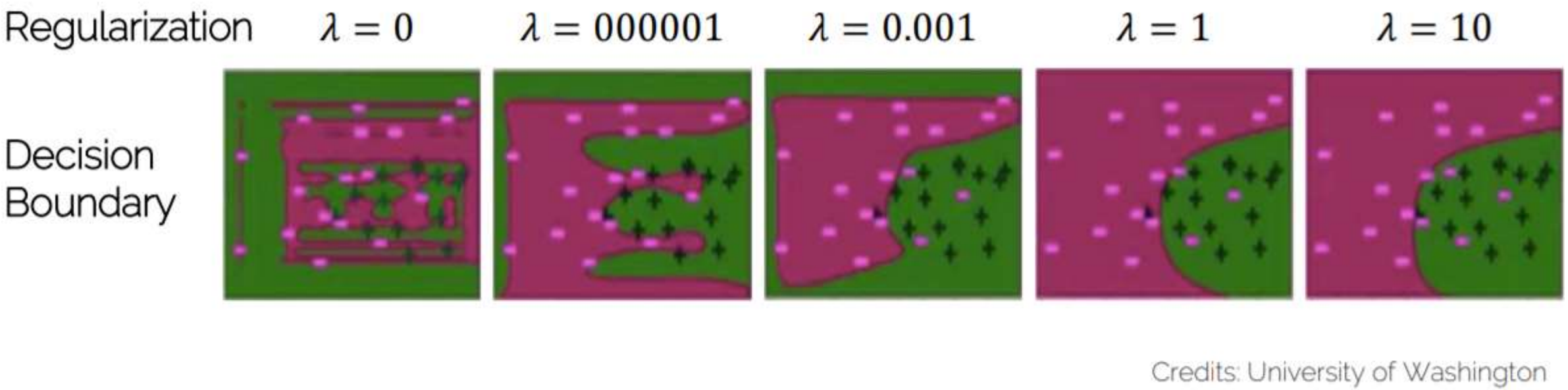


L2 regularization will take all information into account to make decisions

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regularization



Credits: University of Washington

Regularization

- Any strategy that aims to

Source: Niessner and Leal-Taixé, Introduction to Deep Learning